Lab Sheet 04 & 05

Title: Update, Delete Commands in MongoDB.

Objective:

- Understand about Update operations.
- Understand delete commands.

Delete Operations

1. Delete students who have the status "**Graduated**".

```
db.Students.deleteMany({
  Status: "Graduated"
})
```

2. Remove students from **Grade 11** who enrolled **before 2020**.

```
db.Students.deleteMany({
  Grade: 11,
  EnrollmentDate: { $lt: new Date("2020-01-01") }
})
```

3. Delete a specific student record for "**James White**".

```
db.Students.deleteOne({
  Name: "James White"
})
```

4. Remove the "**Scholarship**" field from all students.

```
db.Students.updateMany(
  {},
  { $unset: { Scholarship: "" } }
)
```

Exercise 01

Dataset: Books Collection

BookID	Title	Author	Genre	Published Year	Copies Available	Total Copies	Status
301	The Great Gatsby	F. Scott Fitzgerald	Fiction	1925	3	5	Available
302	To Kill a Mockingbird	Harper Lee	Fiction	1960	1	4	Available
303	1984	George Orwell	Dystopian	1949	0	6	Borrowed
304	The Catcher in the Rye	J.D. Salinger	Fiction	1951	2	3	Available
305	Moby-Dick	Herman Melville	Adventure	1851	5	5	Available
306	Pride and Prejudice	Jane Austen	Romance	1813	1	4	Available
307	The Hobbit	J.R.R. Tolkien	Fantasy	1937	0	7	Borrowed
308	War and Peace	Leo Tolstoy	Historical	1869	2	5	Available
309	Hamlet	William Shakespeare	Drama	1603	4	6	Available
310	Crime and Punishment	Fyodor Dostoevsky	Crime	1866	3	4	Available

1. Create a new database called **LibraryDB.**

```
use LibraryDB
```

2. Create a collection named **Books**.

```
db.createCollection("Books")
```

3. Insert all the book records from the dataset above.

```
db.Books.insertMany([
    {BookID: 301, Title: "The Great Gatsby", Author: "F. Scott Fitzgerald", Genre: "Fiction", PublishedYear: 1925, CopiesAvailable: 3, TotalCopies: 5, Status: "Available"
    },
    {BookID: 302, Title: "To Kill a Mockingbird", Author: "Harper Lee", Genre: "Fiction", PublishedYear: 1960, CopiesAvailable: 1, TotalCopies: 4, Status: "Available"
```

```
    },
    {BookID: 303, Title: "1984", Author: "George Orwell", Genre: "Dystopian", PublishedYear:
1949, CopiesAvailable: 0, TotalCopies: 6, Status: "Borrowed"
    },
    {BookID: 304, Title: "The Catcher in the Rye", Author: "J.D. Salinger", Genre:
"Fiction", PublishedYear: 1951, CopiesAvailable: 2, TotalCopies: 3, Status: "Available"
    },
    {BookID: 305, Title: "Moby-Dick", Author: "Herman Melville", Genre: "Adventure",
PublishedYear: 1851, CopiesAvailable: 5, TotalCopies: 5, Status: "Available"
    },
    {BookID: 306, Title: "Pride and Prejudice", Author: "Jane Austen", Genre: "Romance",
PublishedYear: 1813, CopiesAvailable: 1, TotalCopies: 4, Status: "Available"
    },
    {BookID: 307, Title: "The Hobbit", Author: "J.R.R. Tolkien", Genre: "Fantasy",
PublishedYear: 1937, CopiesAvailable: 0, TotalCopies: 7, Status: "Borrowed"
    },
    {BookID: 308, Title: "War and Peace", Author: "Leo Tolstoy", Genre: "Historical",
PublishedYear: 1869, CopiesAvailable: 2, TotalCopies: 5, Status: "Available"
    },
    {BookID: 309, Title: "Hamlet", Author: "William Shakespeare", Genre: "Drama",
PublishedYear: 1603, CopiesAvailable: 4, TotalCopies: 6, Status: "Available"
    },
    {BookID: 310, Title: "Crime and Punishment", Author: "Fyodor Dostoevsky", Genre:
"Crime", PublishedYear: 1866, CopiesAvailable: 3, TotalCopies: 4, Status: "Available"
    }
])
```

4. Increase the total copies of "**1984**" by **2** and adjust available copies accordingly.

```
db.Books.updateOne(
  {Title:"1984"},
  {
    $inc: {
      TotalCopies:2,
      CopiesAvailable:2
    }
  }
)
```

5. Mark "**Pride and Prejudice**" as "**Reserved" if copies available are less than 2.**

```
db.Books.updateOne(
  {Title:"Pride and Prejudice", CopiesAvailable:{$lt:2}},
  {$set:{Status:"Reserved"}}
)
```

6. Update all books published **before 1900** to have a **new field "Classic**": true.

```
db.Books.updateMany(
  {PublishedYear:{$lt:1900}},
  {$set:{Classic:true}}
)
```

7. Change the **genre of "The Hobbit**" to "**Epic Fantasy**".

```
db.Books.updateOne(
  {Title:"The Hobbit"},
  {$set:{Genre:"Epic Fantasy"}}
)
```

8. Add a "**LastBorrowedDate**" field to books that are currently borrowed (set a sample date).

```
db.Books.updateMany(
  {Status:"Borrowed"},
  {$set:{LastBorrowedDate:new Date("2024-03-01")}}
)
```

9. Delete all books where **TotalCopies** is **less than 3** (outdated books removal).

```
db.Books.deleteMany(
  {TotalCopies:{$lt:3}}
)
```

10. Remove the "**Crime and Punishment**" book from the database.

```
db.Books.deleteOne(
  {Title:"Crime and Punishment"}
)
```

11. Remove the "**Classic"** field from books **published after 1900** (to correct the dataset).

```
db.Books.updateMany(
  {PublishedYear:{$gt:1900}},
  {$unset:{Classic:""}}
)
```

12. Delete books that have been **borrowed but have 0 total copies left**.

```
db.Books.deleteMany(
  {Status:"Borrowed", TotalCopies:0}
)
```

Exercise 02

ProductID	Name	Category	Price (USD)	Stock Quantity	Supplier	ExpiryDate	Discount (%)	Availability
101	Apple	Fruits	1.50	100	Fresh Farms	2025-06-20	5	In Stock
102	Orange Juice	Beverages	3.00	50	Nature's Best	2024-12-10	10	In Stock
103	Rice (5kg)	Grains	12.00	40	Agro Supplies	2026-01-01	0	In Stock
104	Chicken Breast	Meat	8.50	20	Farm Fresh	2024-11-15	15	In Stock
105	Whole Wheat Bread	Bakery	2.50	60	Baked Goods	2024-09-05	5	In Stock
106	Milk (1L)	Dairy	2.00	80	Dairy Fresh	2024-08-20	0	In Stock
107	Olive Oil (1L)	Cooking Oil	10.00	30	Health Foods	2026-05-15	5	In Stock
108	Chocolate Bar	Snacks	1.20	200	Sweet Treats	2025-02-18	20	In Stock
109	Cereal Box	Breakfast	4.50	90	Healthy Start	2025-03-22	10	In Stock
110	Eggs (Dozen)	Dairy	3.50	70	Farm Fresh	2024-07-30	0	In Stock

1. Create a new database called SupermarketDB.

```
use SupermarketDB
```

2. Create a collection named Products.

```
db.createCollection("Products")
```

3. Insert all product records into the Products collection.

```
db.Products.insertMany([
{ProductID:101, Name:"Apple", Category:"Fruits", Price:1.50, StockQuantity:100,
Supplier:"Fresh Farms", ExpiryDate:new Date("2025-06-20"), Discount:5, Availability:"In
Stock"},
```

```
{ProductID:102, Name:"Orange Juice", Category:"Beverages", Price:3.00, StockQuantity:50,
Supplier:"Nature's Best", ExpiryDate:new Date("2024-12-10"), Discount:10, Availability:"In
Stock"},
{ProductID:103, Name:"Rice (5kg)", Category:"Grains", Price:12.00, StockQuantity:40,
Supplier:"Agro Supplies", ExpiryDate:new Date("2026-01-01"), Discount:0, Availability:"In
Stock"},
{ProductID:104, Name:"Chicken Breast", Category:"Meat", Price:8.50, StockQuantity:20,
Supplier:"Farm Fresh", ExpiryDate:new Date("2024-11-15"), Discount:15, Availability:"In
Stock"},
{ProductID:105, Name:"Whole Wheat Bread", Category:"Bakery", Price:2.50, StockQuantity:60,
Supplier:"Baked Goods", ExpiryDate:new Date("2024-09-05"), Discount:5, Availability:"In
Stock"},
{ProductID:106, Name:"Milk (1L)", Category:"Dairy", Price:2.00, StockQuantity:80,
Supplier:"Dairy Fresh", ExpiryDate:new Date("2024-08-20"), Discount:0, Availability:"In
Stock"},
{ProductID:107, Name:"Olive Oil (1L)", Category:"Cooking Oil", Price:10.00, StockQuantity:30,
Supplier:"Health Foods", ExpiryDate:new Date("2026-05-15"), Discount:5, Availability:"In
Stock"},
{ProductID:108, Name:"Chocolate Bar", Category:"Snacks", Price:1.20, StockQuantity:200,
Supplier:"Sweet Treats", ExpiryDate:new Date("2025-02-18"), Discount:20, Availability:"In
Stock"},
{ProductID:109, Name:"Cereal Box", Category:"Breakfast", Price:4.50, StockQuantity:90,
Supplier:"Healthy Start", ExpiryDate:new Date("2025-03-22"), Discount:10, Availability:"In
Stock"},
{ProductID:110, Name:"Eggs (Dozen)", Category:"Dairy", Price:3.50, StockQuantity:70,
Supplier:"Farm Fresh", ExpiryDate:new Date("2024-07-30"), Discount:0, Availability:"In Stock"}
])
```

4. Retrieve all products in the collection.

```
db.Products.find()
```

5. Find details of the product named "Olive Oil (1L)".

```
db.Products.find({Name:"Olive Oil (1L)"})
```

6. Find all products in the "Dairy" category.

```
db.Products.find({Category:"Dairy"})
```

7. Find products that are priced above **$5.00**.

```
db.Products.find({Price:{$gt:5}})
```

8. Find products that have a discount greater than **10%**.

```
db.Products.find({Discount:{$gt:10}})
```

9. Find products with stock quantity below **50**.

```
db.Products.find({StockQuantity:{$lt:50}})
```

10. Find all products expiring before **December 31, 2024**.

```
db.Products.find({ExpiryDate:{$lt:new Date("2024-12-31")}})
```

11. Retrieve all products supplied by "Farm Fresh".

```
db.Products.find({Supplier:"Farm Fresh"})
```

12. Find products where the name starts with "C".

```
db.Products.find({Name:/^C/})
```

13. Increase the price of "Apple" by **$0.50**.

```
db.Products.updateOne(
  {Name:"Apple"},
  {$inc:{Price:0.50}}
)
```

14. Update all products in the "Bakery" category to include "Gluten-Free": false.

```
db.Products.updateMany(
  {Category:"Bakery"},
  {$set:{"Gluten-Free":false}}
)
```

15. Set "Availability": "Out of Stock" for products with stock quantity less than **20**.

```
db.Products.updateMany(
  {StockQuantity:{$lt:20}},
  {$set:{Availability:"Out of Stock"}}
)
```

16. Apply an extra **5% discount** to products in the "Snacks" category.

```
db.Products.updateMany(
  {Category:"Snacks"},
  {$inc:{Discount:5}}
)
```

17. Change the supplier of "Milk (1L)" to "Dairy Best".

```
db.Products.updateOne(
  {Name:"Milk (1L)"},
```

```
  {$set:{Supplier:"Dairy Best"}}
)
```

18. Add a "Rating" field with a default value of **4.0** for all products.

```
db.Products.updateMany(
  {},
  {$set:{Rating:4.0}}
)
```

19. Remove all products that have expired (ExpiryDate before today).

```
db.Products.deleteMany(
  {ExpiryDate:{$lt:new Date()}}
)
```

20. Delete the "Chocolate Bar" product from the collection.

```
db.Products.deleteOne(
  {Name:"Chocolate Bar"}
)
```

21. Remove the "Discount" field from all products in the "Grains" category.

```
db.Products.updateMany(
  {Category:"Grains"},
  {$unset:{Discount:""}}
)
```